

International Journal of Remote Sensing

ISSN: 0143-1161 (Print) 1366-5901 (Online) Journal homepage: www.tandfonline.com/journals/tres20

Cross-temporal domain generalization approach combined with self-organizing-maps for classifying MODIS time series data

Daniel H. Shibuya, Júlio C.D.M. Esquerdo, João P.S. Werner, André S. Tavares & Filipe C. Felix

To cite this article: Daniel H. Shibuya, Júlio C.D.M. Esquerdo, João P.S. Werner, André S. Tavares & Filipe C. Felix (2025) Cross-temporal domain generalization approach combined with self-organizing-maps for classifying MODIS time series data, International Journal of Remote Sensing, 46:15, 5802-5831, DOI: 10.1080/01431161.2025.2527988

To link to this article: https://doi.org/10.1080/01431161.2025.2527988

© 2025 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.

Published online: 13 Jul 2025.

Submit your article to this journal

Article views: 1288

View related articles

View Crossmark data

Citing articles: 1 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=tres20

INTERNATIONAL JOURNAL OF REMOTE SENSING
2025, VOL. 46, NO. 15, 5802–5831
https://doi.org/10.1080/01431161.2025.2527988

Taylor & Francis
Taylor & Francis Group

OPEN ACCESS Check for updates

Cross-temporal domain generalization approach combined with self-organizing-maps for classifying MODIS time series data

Daniel H. Shibuya [a], Júlio C.D.M. Esquerdo [a,b], João P.S. Werner [a], André S. Tavares [c] and Filipe C. Felix [c]

[a]School of Agricultural Engineering (FEAGRI), University of Campinas (UNICAMP), Campinas, Brazil; [b]Embrapa Digital Agriculture, Brazilian Agricultural Research Corporation, Campinas, Brazil; [c]Brazilian National Council for Scientific and Technological Development (CNPq) grantees conducting research at Embrapa Digital Agriculture, Brazilian Agricultural Research Corporation, Campinas, Brazil

ABSTRACT

This study addresses the challenge of limited labelled samples for land use land cover (LULC) classification by proposing a cross-temporal Domain Generalization (DG) approach. A classification model trained on the 2019/2020 crop year was applied to classify 18 crop years using Moderate Resolution Imaging Spectroradiometer (MODIS) time series data. Training samples were derived from the Brazilian Government's TerraClass project, providing reliable LULC reference data for the Cerrado and Amazon biomes in the state of Mato Grosso (MT), Brazil. The Self-Organizing-Maps (SOM) algorithm was used to ensure data quality by characterizing thematic classes, identifying intra-class variability and excluding inconsistent samples. Three classification algorithms were tested: Random Forest (RF), Temporal Convolutional Neural Network (TempCNN) and Lightweight Temporal Attention Encoder (L-TAE). The DG performance of each algorithm was evaluated using independent samples from the 2016/2017, 2020/2021 and 2021/2022 crop years. RF achieved an overall accuracy (OA) of 0.917, outperforming TempCNN (0.886) and L-TAE (0.859), and consistently achieved higher F1-Scores across all thematic classes, particularly Semi-perennial crops (SP) and Pasture (PA). All three algorithms exhibited higher omission and commission errors in the SP and PA classes, which share similar spectrotemporal patterns in the analysed MODIS spectral bands and vegetation indices. Although limitations were identified, RF effectively classified the 18-year time series, identifying LULC changes and intra-class variability over time, thereby demonstrating the potential of DG approach for generating classifications in years without labelled data. This study presents a structured data characterization process using SOM, which played a crucial role in the success of the DG approach. The results and interpretations provide valuable insights and directions for future DG studies, supporting efforts to mitigate the limitations posed by limited labelled data.

ARTICLE HISTORY
Received 30 January 2025
Accepted 25 June 2025

KEYWORDS
Land use land cover; random forest; deep neural networks; sample quality

CONTACT Daniel H. Shibuya d166222@dac.unicamp.br School of Agricultural Engineering (FEAGRI), University of Campinas (UNICAMP), 501 Cândido Rondon Avenue, Campinas 13083-875,Brazil

© 2025 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.
This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial-NoDerivatives License (http://creativecommons.org/licenses/by-nc-nd/4.0/), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited, and is not altered, transformed, or built upon in any way. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

1. Introduction

Machine learning (ML) methods are widely used in remote sensing (RS) due to their capacity to handle vast amounts of data and learning patterns. Among these, algorithms like Random Forest (RF) are utilized to identify land use land cover (LULC) patterns such as agricultural crops, livestock systems and natural vegetation (C. L. M. D. O. Santos et al. 2019; Souza et al. 2020; Werner et al. 2024). In contrast, Deep Learning (DL) methods are being explored in RS, especially in Satellite Image Time Series (SITS) approaches. DL methods are widely used for complex tasks like image classification and text generation, and their ability to extract and learn features offers potential for RS applications (Fu et al. 2024).

In this study, three classifiers were evaluated: RF, Temporal Convolutional Neural Network (TempCNN) and Lightweight Temporal Attention Encoder (L-TAE). RF was selected over other ML algorithms, such as Support Vector Machine (SVM), due to its well-known effectiveness in LULC classification tasks and adaptability to time series data (Luciano, Campagnuci, and le Maire 2022; Pelletier et al. 2016; Xu et al. 2020). Moreover, RF models, which are based on the decision tress, offer greater interpretability and are less sensitive to imbalance data (Belgiu and Drăgu 2016; Breiman 2001). Unlike ML approaches, which treat time series as tabular data, DL methods are able of capturing temporal dependencies and correlations within sequential data (Ismail Fawaz et al. 2019; Santosh, Das, and Ghosh 2022). TempCNN and L-TAE are neural network-based algorithms designed for time series analysis, as highlighted by Pelletier, Webb, and Petitjean (2019) and Garnot and Landrieu (2020), respectively. However, DL algorithms involve a complex learning process that occurs across multiple hidden layers, making them less interpretable than ML algorithms such as RF. Additionally, both ML and DL methods typically require a larger number of samples to effectively train the model, and overcome issues such as overfitting (Y. Wang et al. 2023).

In RS, training and validating models to map large regions, such as biomes, typically relies on ground samples representing the diversity of thematic classes. However, acquisition of ground samples through field surveys is challenging and expensive, and many regions lack sufficient labelled data for studies (Feng et al. 2024; Hao et al. 2020). In this context, methodologies for addressing these challenges contribute to expanding the scope of research in RS. In this paper, we utilized TerraClass maps as an alternative data source for acquiring training samples. TerraClass Project (www.terraclass.gov.br.) is a partnership between the Brazilian Agricultural Research Corporation (Embrapa) and the National Space Research Institute (INPE) to produce the Brazilian Government's official LULC data for two biomes: Amazon and Cerrado (Brazilian savannah) (Almeida et al. 2016). TerraClass utilizes Sentinel-2/MSI (Multispectral Instrument) data with 10-metre spatial resolution data and includes 16 thematic classes designed to monitor changes in natural vegetation and agriculture expansion, with overall accuracy (OA) of 0.85 for the Amazon and 0.88 for the Cerrado.

Samples derived from mapping projects can help mitigate the scarcity of ground samples; however, these samples are restricted to the specific spatial extent, reference year of the map and may contain a certain degree of uncertainty. Consequently, some regions and time periods still lack sufficient samples for training models. To address this issue, approaches such as Domain Generalization (DG) and Domain Adaptation (DA) have been explored (Mohammadi, Belgiu, and Stein 2024; Zhu et al. 2023). These approaches involve training a base model in a well-sampled domain

(Source Domain (SD)) and subsequently applying to target areas lacking labelled samples (Target Domain (TD)), differing from the traditional approach, where a ML or DL classification model is trained and tested within a specific study area using a designated dataset (Feng et al. 2024; Xu et al. 2020; Zhou et al. 2021).

Despite their similarity, DA and DG have distinct methodologies. DA incorporates samples from the TD during the training process, minimizing differences between SD and TD by adapting the model to new data. In contrast, DG is designed to develop models with inherent robustness, enabling them to generalize to unseen TD data (J. Wang et al. 2021; Zhu et al. 2023). This characteristic emphasizes the potential of DG for RS applications but also distinguishes it from other approaches, as it facilitates the continuous generation of classification maps across distinct time periods and regions, eliminating the need of retraining or incorporating additional samples (Luciano, Campagnuci, and le Maire 2022; Mohammadi, Belgiu, and Stein 2024).

For instance, Massey et al. (2017) applied cross-temporal DG to address the need of rapid and reliable long-term cropland monitoring in the United States. Their model adapted Cropland Data Layer (CDL) reference data to Moderate Resolution Imaging Spectroradiometer (MODIS) 250 m spatial resolution SITS data, aiming to map agricultural crops such as cotton and corn. DG model results were similar to those of year-specific models in 2008, 2010, 2011 and 2012, with less than 3% of difference in OA. Furthermore, the authors highlight the potential of cross-temporal DG in enabling the generation of agricultural crops maps from MODIS data for the period 2001 to 2007, a period which such maps were previously unavailable. Luciano, Campagnuci, and le Maire (2022) also utilized cross-temporal DG and performed a 33-year classification of sugarcane in São Paulo, Brazil. Their approach achieved the best results using a RF model trained on samples from a six-year period (2009–2014), resulting in an average F1-Score of 0.83 and 0.8 for test years 2015 and 2016, respectively. Moreover, their 33-year classification demonstrated a high correlation with reference base maps such as MapBiomas (Souza et al. 2020) and data from the Brazilian Institute of Geography and Statistics (IBGE), further demonstrating the robustness of DG for cross-temporal classification.

Although DG holds significant potential, its application remains unexplored due to inherent implementation challenges. Consequently, many studies relied on the DA approach to mitigate scarcity of data (Ge et al. 2021; Hao et al. 2020). In RS, generalization challenges arise in cross-regional and cross-temporal applications due to differences in the spectrotemporal characteristics of thematic classes across different years and regions (Mohammadi, Belgiu, and Stein 2024). As a result, developing a model capable of adapting to diverse, unseen data remains a challenge for ML and DL, as such a model may lose effectiveness and lead to inaccurate results (Blanchard, Lee, and Scott 2011; J. Wang et al. 2021).

To effectively implement DG, Zhong, Gong, and Biging (2014) and Luciano et al. (2018) emphasize the importance of maintaining consistency between the thematic classes in the SD and TD, and of accounting for intra-class variability in the classification model when applied to unseen data. Moreover, they highlight the need of a detailed process of training sample characterization to ensure that the representative properties of each thematic class are accurately captured and informed to the model. This variability can arise from factors such as differing crop calendars, climate fluctuations, and regional management practices, which impact the spectrotemporal characteristics of the same thematic classes across various locations and time periods (Johnson 2019; Khanna et al. 2022).

To address these challenges, this study applied the Self-Organizing-Maps (SOM) algorithm to identify intra-class variability and characterize samples by evaluating their quality and detecting noise within the dataset (L. A. Santos, Ferreira, Picoli, et al. 2021). SOM was selected over traditional clustering techniques, such as hierarchical clustering and K-means, due to its compatibility with time series data and its ability to provide an interpretable process for detecting sample inconsistencies. L. A. Santos, Ferreira, Camara, et al. (2021) utilized SOM to identify intra-class variability, their approach identified ten distinct patterns within double-crop thematic class across the Cerrado biome and two patterns within the pasture class. Additionally, L. A. Santos, Ferreira, Camara, et al. (2021) demonstrated that SOM-based quality assessment improved the performance of a RF classifier by 4% in OA, increasing it from 94.3% to 98.4%.

Therefore, this study aimed to address the scarcity of training samples by implementing a sampling strategy based on the TerraClass mapping and applying a cross-temporal DG approach. TerraClass maps were selected as they constitute the official Brazilian dataset for monitoring LULC in the Cerrado and Amazon biomes, providing a reliable source of reference data. To ensure quality, sample characterization and identification of intra-class variability was performed using SOM. This methodology was employed to classify LULC thematic classes, including agricultural crops and natural vegetation, in an interannual classification of MODIS SITS data. We hypothesize that a classification model trained on a robust and well-characterized dataset can maintain its performance across multiple years of MODIS time series, reducing the need for frequent adaptation and retraining.

2. Material and methods

2.1. Study area

The study area, Mato Grosso (MT), one of Brazil's largest states (903,357 km^2), spans portions of the Cerrado (Savannah), Amazon and Pantanal biomes (Figure 1). This region holds significant agricultural importance for Brazil, being the country's main producer of grains and fibbers, especially soybean, corn, and cotton. During the summer season (October to February), soybean is the predominant crop, while in the off-season (April to July), the main crops are corn, cotton, sorghum and sunflower (Dantas Chaves and Alves 2019). This double-cropping system is well established in MT and allows grain production in two distinct periods. Moreover, with technological advancements, some regions exhibit three production cycles per crop year (Vieira et al. 2022). Additionally, non-commercial crops, such as crotalaria and millet, are often cultivated in rotation with commercial crops to promote conservation management practices (Arvor et al. 2012). Notably, areas with single cropping systems are also observed in the study area, together with extensive sugarcane fields and pasture areas. In the Amazon biome, dense forest natural vegetation prevails, characterized by its typical biodiversity and ecological importance.

Mato Grosso was selected as the study area for applying the DG approach due to its diverse range of natural vegetation types and agricultural crops distributed across the state. Additionally, the region exhibits a complex interplay between agricultural expansion and conservation efforts. Over the years, agricultural intensification has led to the transition from single-cropping to double-cropping systems, while conservation policies,

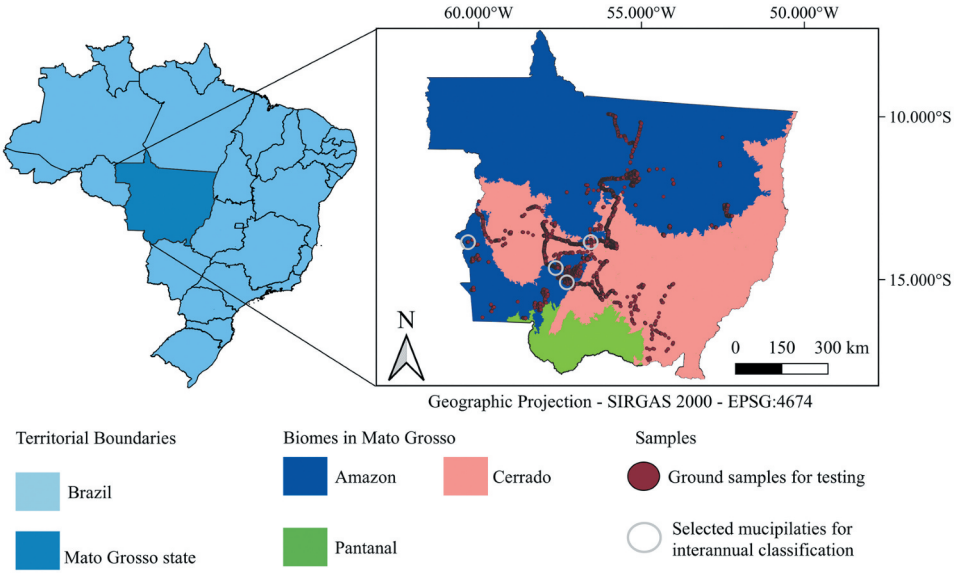


Figure 1. Map of the Brazilian state of Mato Grosso (MT), portions of its three biomes, distribution of ground samples for testing and localization of four municipalities with selected samples for testing interannual classification.

particularly in the Amazon biome, have aimed to balance economic development with environmental preservation (Vieira et al. 2022). This diversity in LULC provides variability across different crop years, making it an appropriate region for evaluating the model's generalization capability.

The study followed the Brazilian agricultural calendar, defining the crop year as the period from September of one year to August of the following year, according to the Brazilian National Supply Company (CONAB), encompassing both summer season and off-season cycles in Mato Grosso (CONAB 2021).

2.2. Overall methodology

The workflow adopted in this study is outlined in detail in Figure 2. Training samples were derived from the 2020 version of the TerraClass maps, with each sample assigned its corresponding time series data from the MODIS sensor. The original set of samples was referred to as Original sampling (OS), and was used to examine the spectrotemporal patterns of each thematic class as a preliminary step for sample characterization. OS was also used as the input for the SOM algorithm, which was applied to identify and remove inconsistent samples. The resulting dataset, after SOM treatment, was referred to as SOM Sampling (SS). Classification models were trained using SS and three algorithms: Random Forest (RF), Temporal Convolutional Neural Network (TempCNN) (Pelletier, Webb, and Petitjean 2019) and Lightweight Temporal Attention Encoder (L-TAE) (Garnot and Landrieu 2020). Lastly, the trained models were applied to the cross-temporal DG approach using independent ground samples for testing.

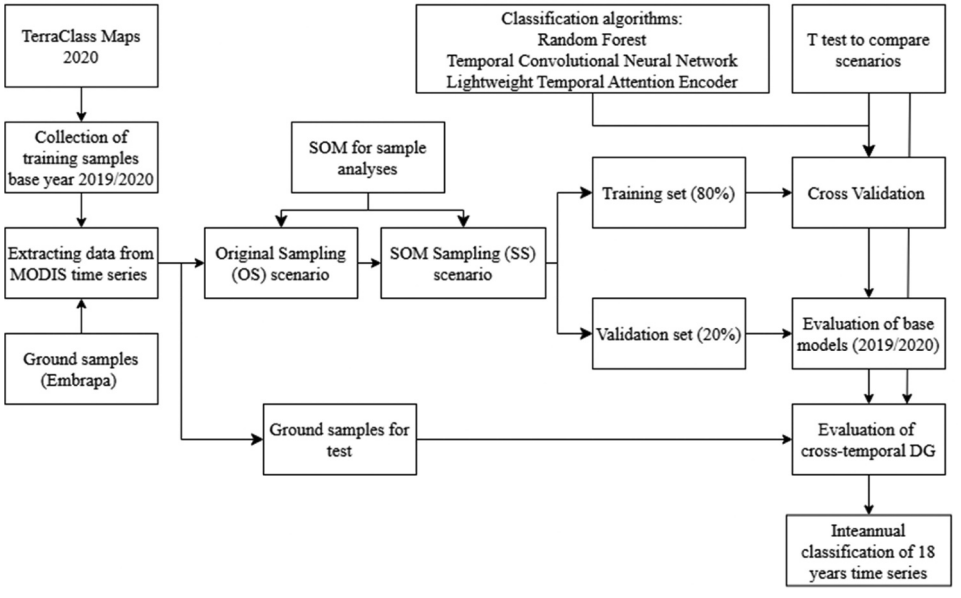


Figure 2. Flowchart illustrating the overall methodology used in this study. DG: domain generalization. SOM: self-Organizing-maps.

2.3. MODIS data

We selected MODIS products based on the agricultural production patterns in Mato Grosso and the compatibility of their 250-metre resolution with previous studies of LULC in this region (Esquerdo et al. 2020; Simoes et al. 2020; Y. Wang et al. 2023; Zhang et al. 2021). Mato Grosso is characterized by extensive agricultural fields, often ranging from 30 to over 1000 hectares, with the largest fields mainly dedicated to corn and soybean cultivation (Brown et al. 2013). Additionally, MODIS provides a continuous time series, making it a valuable dataset for evaluating the generalization capabilities of classification models across several crop years. In this study, we utilized an 18-year time series (2004 to 2022) of 16-day composite products with radiometric, geometric and atmospheric correction, derived from MOD13Q1 and MYD13Q1, corresponding to the TERRA and AQUA satellite collections, respectively (Justice et al. 2002). These composites are generated by selecting the highest radiometric and geometric quality pixels within a 16-day period, prioritizing observations with minimal cloud contamination and optimal viewing angles.

To ensure a systematic analysis of SITS data, we processed the MODIS collection obtained from the Land Process Distributed Active Archive Center (LP DAAC) Data Pool repository (https://lpdaac.usgs.gov). The processing involved organizing the data into image stacks containing essential layers, such as NDVI (Normalized Difference Vegetation Index) (Rouse et al. 1973), EVI (Enhanced Vegetation Index) (Huete, Liu, and Wim 1997) and spectral bands from MODIS products. To facilitate interpretation, these spectral bands were labelled as BLUE, RED, NIR1 and NIR2, according to their respective wavelengths (Table 1). Additionally, the data stack included a CLOUD band, derived from the pixel reliability layers available in MOD/MYD13Q1 products. This layer provides

Table 1. Labelled spectral bands from MODIS product. Source: (https://lpdaac.usgs.gov.).

Label	MODIS spectral band	Reflected range (nm)
BLUE	Band 3 (Visible Spectrum)	459–479
RED	Band 1 (Visible Spectrum)	620–670
NIR1	Band 2 (Shortwave Infrared)	841–876
NIR2	Band 5 (Shortwave Infrared)	1230–1250

information on data quality, especially cloud presence. To address gaps caused by cloud cover, a bilinear interpolation method was applied, replacing missing values and ensuring the continuity of the time series (Esquerdo et al. 2020).

2.4. Training samples

2.4.1. Base maps and thematic classes

In this study, we addressed the scarcity of labelled samples by utilizing TerraClass as the base maps for acquiring training samples. We selected the maps from the 2020 version, corresponding to the crop year 2019/2020 in Mato Grosso.

TerraClass maps includes 16 thematic classes, which some of them were grouped into five general class types related to agricultural crops and natural vegetation, as detailed in Table 2. Thematic classes representing natural vegetation in TerraClass were consolidated to broader categories named Pasture (PA) and Forest Vegetation (FV). This adaptation was necessary because thematic classes, such as forestry and shrub pasture, which function at the crop level, might be underrepresented in the samples. This potential under-sampling could pose challenges for ML and DL algorithms, so these classes were grouped with others whose spectrotemporal patterns are similar over the course of a single crop year. Thematic classes representing agricultural crops adhered to the standard used in TerraClass and did not require any adaptations.

In this study, the expected spectrotemporal patterns of each thematic class are based on the TerraClass description (Table 2) and NDVI profile characterizations reported in previous studies in the same region (Picoli et al. 2018; Simoes et al. 2020; Vieira et al. 2022; Zheng et al. 2022). Semi-perennial crops (SP), primarily represented by sugarcane, exhibit

Table 2. Thematic classes and their respective descriptions available in TerraClass maps, which served as the basis for the five thematic classes used in this study. Source: TerraClass (www.terraclass.gov.br.).

Thematic classes	TerraClass classes	Description
Single Crop (SC)	Single crop	Agricultural crops that have a single production cycle in the crop year.
Double Crop (DC)	Double crop	Agricultural crops that have two or more production cycles in the crop year.
Semi-perennial Crop (SP)	Semi-perennial crop	Agricultural crops with a cycle spanning more than one crop year, primarily exemplified by sugar cane.
Pasture (PA)	Herbaceous pasture	Pastures with a predominance of herbaceous forage vegetation.
	Shrub pasture	Pastures with a predominance of woody vegetation, made up of shrub/tree species.
Forest Vegetation (FV)	Primary forest vegetation	Natural vegetation characterized by the densification of tree species that have not suffered from human action.
	Secondary forest vegetation	Natural vegetation in the regeneration process.
	Forestry	Cultivation of forest species of commercial interest, e.g. Eucalyptus and Pinus.

Table 3. Number of samples for each thematic class in original sampling (OS).

Thematic Class	Original Sampling
Double Crop (DC)	10000
Single Crop (SC)	5000
Semi-perennial Agriculture (SP)	5000
Pasture (PA)	10000
Forest Vegetation (FV)	10000
Total	40000

a regular NDVI profile lasting 6 months or more, beginning in the summer season (October–December) and ending in harvest period (April–August). PA areas display NDVI profiles similar to SP but with an irregular pattern and generally lower NDVI values. FV maintains a profile with high and stable NDVI values throughout the entire crop year. Single Crop (SC) present an unimodal NDVI profile lasting approximately three months, while Double Crop (DC) are characterized by a regular and bimodal NDVI profile over the crop year. These NDVI profile descriptions served as a reference for analysing the spectro-temporal patterns in our dataset.

2.4.2. Training sample extraction

TerraClass maps are based on Sentinel-2/MSI data with 10 metres spatial resolution, while MODIS data resolution is 250 metres. To ensure compatibility with MODIS, we adjusted the resolution of TerraClass maps using QGIS (version 3–32) raster projection tools. To mitigate edge effects during sampling, we applied the erode filter, which removes the outer edges of pixel groupings with the same thematic class, focusing the sampling process on the inner pixels. An erosion 'cut-off' distance of 2 pixels (approximately 500 metres) was applied.

Sampling was conducted using the Accuracy Assessment of Thematic Mapas (Acatama) plugin in QGIS (Llano 2023). Following the guidelines of Olofsson et al. (2014), sampling points were randomly distributed across the entire study area to ensure coverage and avoid concentration in specific regions. This process employed a pixel-based approach, where a single sample was collected from each pixel. Each sample included the pixel's geographical coordinates (latitude and longitude) and its corresponding thematic class. Table 3 shows the number of samples for each thematic class.

2.5. Ground samples for testing

To assess and validate the DG results, we used ground samples provided by Embrapa, collected through field campaigns across the entire state of Mato Grosso during the 2016/2017, 2020/2021 and 2021/2022 crop years (Figure 1). This dataset was used to evaluate model's performance on unseen data and determine which classification model achieved better generalization across different crop years. The distribution of the number of samples in each thematic class and crop year are shown in Table 4.

Notably, the ground samples are unbalanced across all crop years. For instance, no FV samples were available for the 2020/2021 crop year, and DC samples were mostly concentrated in 2021/2022. Despite this imbalance, the samples were collected from crop years different from the 2019/2020 training period, aligning with the objective of

Table 4. Number of ground samples for testing of each thematic class.

Thematic Class	2016/2017	2020/2021	2021/2022	Total
Double Crop (DC)	2	3	1623	1628
Single Crop (SC)	74	27	975	1076
Semi-perennial Agriculture (SP)	162	567	176	905
Pasture (PA)	616	7	542	1165
Forest Vegetation (FV)	295	-	459	754
Total	1139	604	3775	5528

the study to characterize the cross-temporal DG approach and evaluate model's generalization capability. Additionally, to further assess the interannual classification, we selected four ground sample points located in different municipalities within the Cerrado and Amazon biomes. These points were selected due to their dynamic agricultural and natural vegetation LULC changes across the 18-year time series, such as transitions between natural vegetation and agriculture thematic classes. This selection aimed for providing valuable data for analysing the adaptability and generalization capacity of the classification models. The locations of the municipalities corresponding to these selected samples are shown in Figure 1.

2.6. Computational package sits

In this study, the main computational tool was the 'sits' (Satellite Image Time Series Analysis on Earth Observation Data Cubes), an open-source R package for satellite image time series analysis. It enables users to apply machine learning techniques for classifying image time series obtained from Earth observation data cubes. This package was introduced by Simoes et al. (2021) and is tailored for handling Earth observation data in a structured manner, presenting functions designed for the LULC classification process using SITS. In this study, following the extraction of the time series data from MODIS image stacks, all processes related to the training classification models, analysing samples with SOM, and evaluating results were performed using functions from the sits package.

2.7. Assessing sample quality using self-organizing-maps (SOM)

Initially proposed by Kohonen (1990) SOM architecture utilizes Competitive Learning for clustering samples through unsupervised learning. This algorithm employs a neural network generating a two-dimensional distribution where data with similar patterns are grouped together. The distribution is composed of units called neurons, which are initialized with random values and are iteratively updated through Competitive Learning. During training, time series with similar spectral patterns are assigned to the same or neighbouring neurons. As training progresses, the neurons organize across the two-dimensional map, forming clusters based on pattern similarity. Some neurons may not receive any assigned sample, indicating patterns not represented in the dataset (L. A. Santos, Ferreira, Camara, et al. 2021).

To address inconsistent samples, SOM calculates two parameters: Conditional Probability (CP) and Posterior Probability (PP), as illustrated in Figure 3. CP reflects the initial likelihood of a sample belonging to a particular neuron. Samples that fall below the designated CP threshold are automatically excluded from the dataset. The PP recalculates

the CP using Bayesian Inference, which accounts for the influence of neighbouring neurons. If the PP fails to meet the minimum threshold, the algorithm indicates that the assignment of the neuron should be reviewed. The decision to exclude or retain the samples is left to the user.

In this study, the SOM was configured following recommendations from sits package documentation and L. A. Santos, Ferreira, Camara, et al. (2021). The input to the SOM algorithm consisted of spectrotemporal data from vegetation indices and spectral bands across all thematic classes. The training involved 100 iterations with a decreasing learning rate from 0.5 to 0.01. Euclidean Distance was used to calculate neuron distance and a 30 × 30 grid (900 neurons) was implemented. A threshold of 60% was applied for CP and PP.

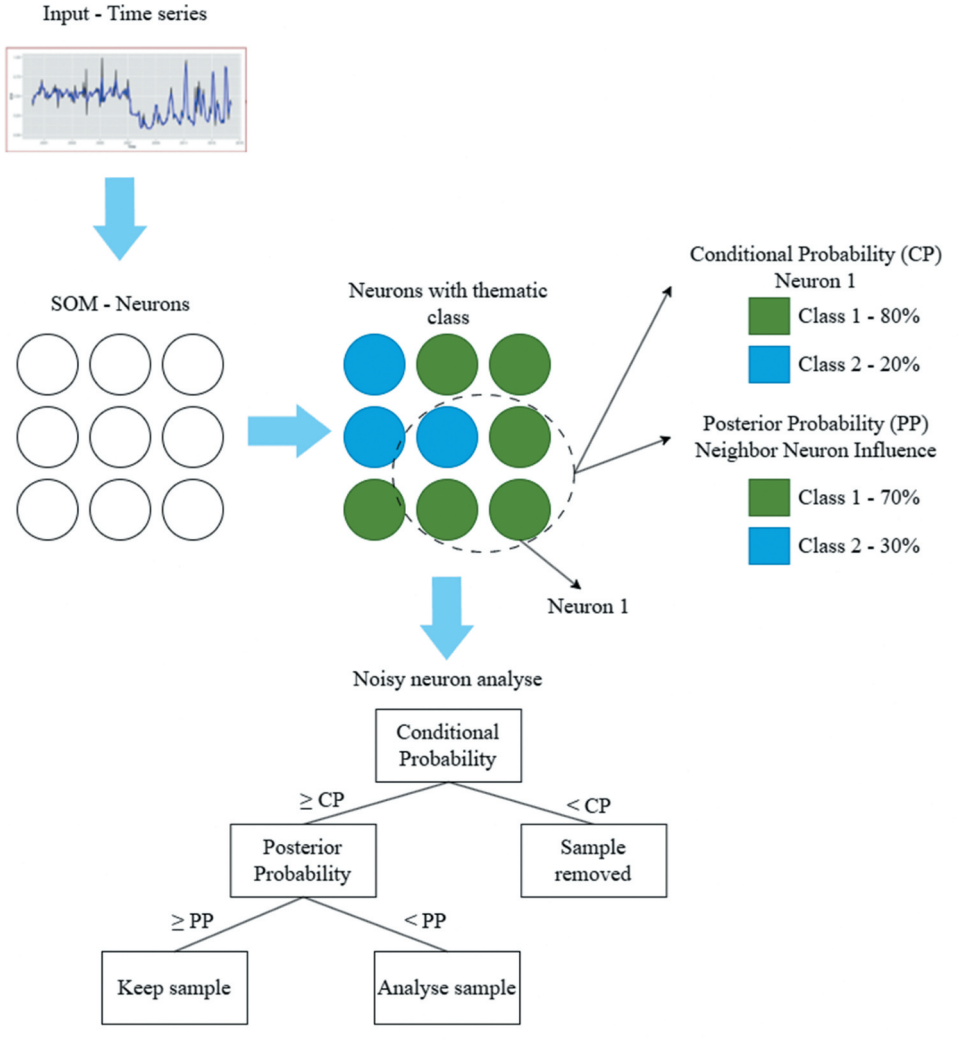


Figure 3. Flowchart exemplifying the two-dimensional distribution creation process and noise analysis methodology using the conditional probability (CP) and posterior probability (PP) metrics. Source: L. A. Santos, Ferreira, Camara, et al. (2021).

In this study, the SOM algorithm was applied in two stages. First, the original set of samples (OS) was used as input to identify inconsistencies within thematic classes. The resulting dataset, with inconsistent samples removed, was referred to as SOM sampling (SS). In the second stage, SOM was applied to the SS to assess intra-class variability and characterize the data that would subsequently be used to train the classification models. For consistency and comparative analysis, the same SOM parameters and a 30×30grid were used in both stages.

2.8. Training of algorithms and validation

In this study, three algorithms were tested. The first one was the Random Forest (RF) (Breiman 2001), an ensemble method based on decision trees that conducts multiple decisions using randomly selected subsets of training samples and variables. RF was defined as a baseline model to evaluate the performance of the other algorithms. To train RF, two main parameters were defined: the number of trees (ntree), set to 500, and the number of variables considered at each split of the decision tree (mtry), set to the square root of the number of input variables.

We also tested Temporal Convolutional Neural Network (TempCNN); a deep neural network proposed by Pelletier, Webb, and Petitjean (2019) specifically designed for handling SITS data focusing on the temporal aspects of the spectrotemporal profiles. Their architecture includes three 1D convolutional layers, each with 64 units, aiming to simplify the propagation of information through the network, reducing the complexity and biases. After passing through these convolutional filters, the processed series are fed into a series of dense layers with 256 units, followed by a softmax layer for classification.

Lastly, we evaluated the Lightweight Temporal Attention Encoder (L-TAE), an adaptation of the Transformer architecture proposed by Garnot and Landrieu (2020). L-TAE utilizes self-attention and positional encoding mechanisms to interpret and classify data. The self-attention mechanism allows the model to discern patterns within the time series, while the positional encoder organizes the temporal information, enabling the classifier to understand the nature of the data.

In deep neural networks, optimizing performance requires tuning a combination of hyperparameters and model parameters (J. Wang et al. 2022). To streamline this process, we adopted the approach recommended by Simoes et al. (2021) for training the TempCNN and L-TAE. Instead of exhaustively testing each possible combination of hyperparameters, we utilized a tuning stage with the AdamW optimizer (Kingma and Ba 2014) and standard parameters as specified in the 'sits_tuning' function from the sits package. The main hyperparameters included $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay set to $1x10^{-6}$, epsilon (ε) = $1x10^{-8}$, learning rate = 0.001, with 150 epochs and a batch size of 64.

We utilized SS to train the classification algorithms, adopting an 80/20 split for training and validation. The split was stratified to ensure proportional representation of all thematic classes. Three classification algorithms (RF, TempCNN and L-TAE) were trained resulting in three different classification models. To statistically compare model performance, we applied a k-fold cross-validation with five repetitions on the training set. Assuming equal variances and a 95% confidence interval, a t-test was conducted on the overall accuracies to assess significant differences among the models (Gerald 2018).

2.9. Cross-temporal domain generalization

The initial evaluation of cross-temporal DG classification was conducted using ground samples as test data (Table 4), representing an unseen dataset for the classification model. Each model performance was assessed using OA and F1-Score, the harmonic means of Precision and Recall, which accounts for both omission and commission errors, allowing the evaluation of classification results for each thematic class individually (Congalton 1991). To statistically compare the DG results, each model was run five times, and a paired t-test was conducted using the OA values. The top-performing classification model identified from this evaluation was subsequently applied to the cross-temporal DG approach and to perform interannual classification over an 18 crop-year period using MODIS SITS data.

To assess LULC changes over time, we selected four ground samples. Due to the scarcity of ground samples for all crop years, our validation primarily relied on visual analyses, supported by findings from previous studies using similar approaches (Picoli et al. 2018; Simoes et al. 2020; Souza et al. 2020; Vieira et al. 2022), as well as on the probability layers generated from the classification results in the sits package. These probability layers offer insights into the likelihood of each sample belonging to specific thematic classes, serving as a valuable tool for identifying potential uncertainties in the classification process and highlighting areas that may require further refinement or investigation (Simoes et al. 2021).

3. Results

3.1. Thematic classes spectrotemporal patterns

To support subsequent SOM analysis, Figure 4 illustrates the spectrotemporal patterns of the average values of vegetation indices and spectral bands for the five thematic classes, based on OS. The expected NDVI profile for each thematic class, described in Section 2.4.1, were used as reference to interpret the patterns show in Figure 4. Detailed spectrotemporal patterns for each thematic class are provided in the supplementary material. Notably, the Single Crop (SC) class did not exhibit the expected pattern characteristic of a single crop cycle, indicating a significant intra-class variability. Additionally, the Semi-perennial Crop (SP) and Pasture (PA) thematic classes displayed similar spectrotemporal patterns, which may pose challenges for LULC classification. On the other hand, the patterns observed for Double Crop (DC) and Forest Vegetation (FV) classes were aligned with their typical behaviour over a single crop year.

3.2. Training samples quality – SOM

3.2.1. Original sampling (OS) analysis

Figure 5 illustrates the two-dimensional distribution generated by applying the SOM algorithm to the OS data, showing groupings of neurons corresponding to the five thematic classes. Neurons exhibiting patterns without any assigned samples were labelled as Not Available. The DC and FV classes are well clustered and spatially distant from each other, indicating distinct and consistent spectrotemporal patterns

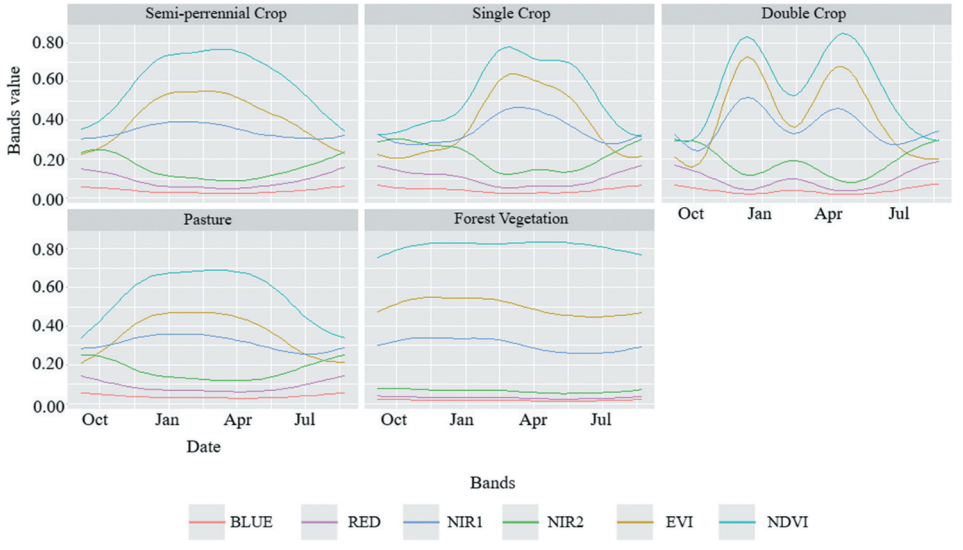


Figure 4. Original sampling (OS) patterns: average values of vegetation indices (NDVI and EVI) and spectral bands (BLUE, RED, NIR1 and NIR2) for the five thematic classes.

unique to each class. In contrast, the SC class has two distinct clusters on the SOM distribution, which aligns with the absence of a well-defined spectral pattern in Figure 4. This dual clustering suggests substantial intra-class variability within the SC class.

The SP class forms two close clusters on the SOM distribution. Additionally, several neurons were observed near those of the PA class, indicating similarities or overlaps in spectrotemporal patterns, as previously noted in Figure 4. The PA class is grouped in proximity to all other thematic classes, particularly SP, suggesting that its patterns may overlap with those of other classes. This proximity makes the PA class more susceptible to misclassification or confusion with neighbouring classes.

Figure 5 illustrates neurons removed from the dataset due to inconsistencies. For example, neuron 151 (Figure 5 (II)) was distant from its respective cluster and had CP of less than 60%, resulting in its automatic exclusion by the algorithm. On the other hand, neuron 791 (Figure 5 (I)) was assigned for further analysis using the PP metric. Upon examining the spectrotemporal patterns, it was determined that neuron 791 did not exhibit the expected characteristic of the SP class. As a result, the samples associated with this neuron were excluded. Minor variations in spectrotemporal profiles were accepted, as they were considered to reflect legitimate intra-class variability. Table 5 summarizes the results from SOM-based analysis, showing that approximately 20% of the OS dataset (8,188 samples) were excluded.

3.2.2. Post-SOM samples (SS) analysis

The second SOM-based distribution was generated from the SS dataset to evaluate the training samples after the removal of inconsistencies and to assess intra-class variability

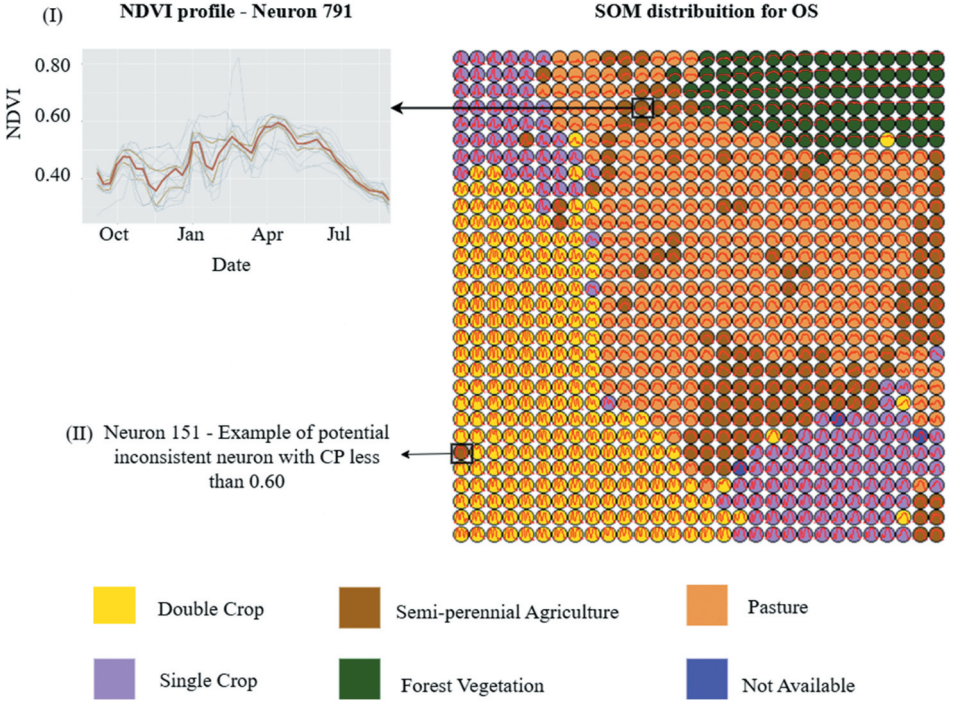


Figure 5. Self-Organizing-maps (SOM) of original samples (OS), highlighting: (I) a potential noisy neuron for semi-perennial agriculture (SP) and their respective NDVI profile for 2019/2020 crop year; and (II) a potential noise neuron with a conditional probability (CP) less than 0.6, indicating a neuron automatically excluded by SOM.

Table 5. Number of samples of each thematic class excluded by SOM.

Thematic class	Original sampling (OS)	SOM sampling (SS)
Double Crop (DC)	10000	8066
Single Crop (SC)	5000	1966
Semi-perennial Agriculture (SP)	5000	4289
Pasture (PA)	10000	8025
Forest Vegetation (FV)	10000	9466
Total	40000	31812

(Figure 6). From this analysis, key findings have emerged: (I) SOM successfully removed inconsistent samples, as demonstrated by the examples in Figure 5; (II) the persistent similarity between PA and SP classes is an inherent characteristic of their spectrotemporal pattern, considering MODIS time series data; (III) FV and DC classes showed a well-defined characteristic across both distributions; and (IV) intra-class variability was identified across all thematic classes, especially in SC and SP.

Additionally, Figure 6 illustrates an example of intra-class variation within the SC. The highlighted neurons exhibited characteristics of SC but differ in their sowing and harvesting periods. Neuron 873 (Figure 6(I)) represents a single crop cycle with late sowing, starting in January of the second year. In contrast, Neuron 150 (Figure 6(II)) also reflects a single crop cycle, but with early sowing at the end of the first year. The observed

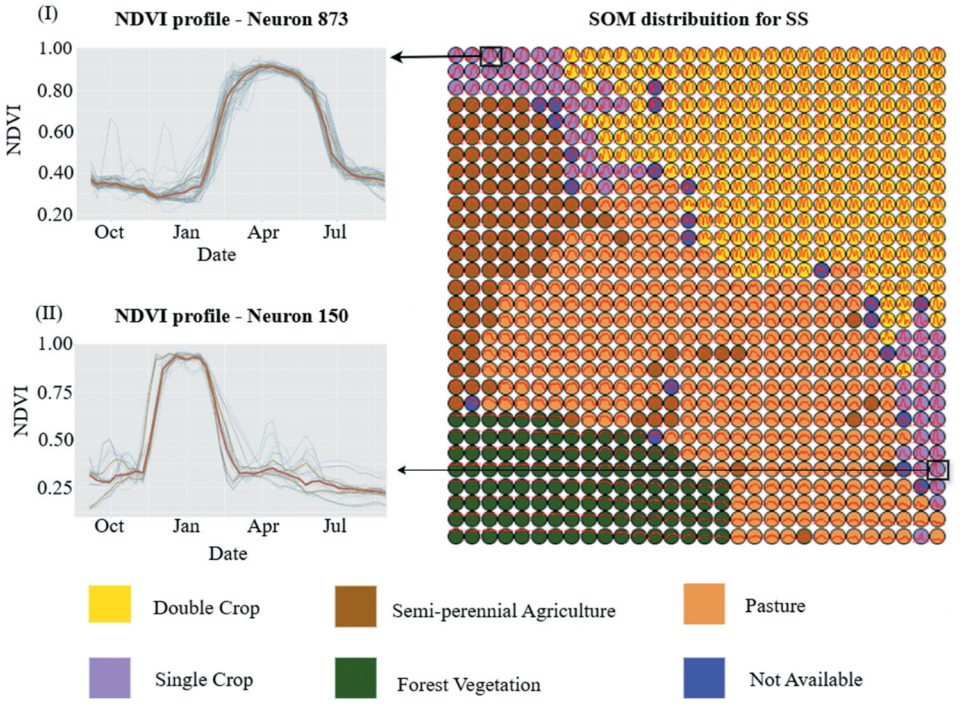


Figure 6. Self-organizing-maps (SOM) of SOM sampling (SS), highlighting two examples of NDVI profiles from neurons 873 (I) and 150 (II), that represents two distinct groups of single crop (SC) thematic class.

differences in Neuron 150 and 873 are associated with the presence of distinct crop types in Mato Grosso. Comparisons with previous studies suggest that neuron 150 exhibits a pattern of corn or soybean, while neuron 873 is more closely aligned with cotton (Oldoni et al. 2019; Simoes et al. 2020). Minor intra-class variability was also detected, such as the occurrence of three crop cycles in DC and natural vegetation in the growing stage in FV. However, these variations had less impact when compared to the variability observed within SC. Examples of the corresponding NDVI profiles for DC and FV are provided in Appendix A.

3.3. Results of the 2019/2020 base model

Base model's results were trained with 80% (25449) subset of the SS dataset, while the results were derived from the 20% (6363) subset separated for testing the model within the same crop year. Table 6 presents the performance of the three algorithms in terms of overall accuracy (OA). All algorithms achieved high OA values, exceeding 0.96 (F1-score and confusion matrices available in Appendix B). Furthermore, statistical analysis confirmed significant differences in their performance, with p-values below 0.05.

Despite the statistical differences, all classification models demonstrated a strong adaptation to the samples from the 2019/2020 crop year. This can be attributed to the fact that the validation set samples share similar characteristics with those used for

Table 6. Overall Accuracy (OA) of each algorithm on the validation set. *indicates statistically significant difference compared to other models (p-value < 0.05).

	Random Forest (RF)	TempCNN	L-TAE
Overall Accuracy (OA)	0.994*	0.981*	0.962*

training, as both are derived from the TerraClass dataset and correspond to the same crop year. Additionally, F1-Score values were consistent across all thematic classes (Appendix B). To further evaluate the classification models' effectiveness, we performed a test using independent ground samples from different crop years.

3.4. Results of the cross-temporal domain generalization

3.4.1. Test with ground samples

Table 7 presents the OA results achieved by each classification model using the ground samples for testing, all samples detailed in Table 4 were used to generate the results of cross-temporal DG. Compared to the results in Table 6, all algorithms had lower values of OA. Among them, RF had the best performance, achieving an OA of 0.917. Additionally, paired t-test confirmed significant differences between the models, with p-values below 0.05.

In the DC, SC and FV thematic classes, all algorithms achieved high F1-Score values (Table 8). As indicated during SOM analysis, these classes exhibit distinct spectrotemporal characteristics, forming with well-defined clusters. Although intra-class variations were observed in SC and DC, all models effectively captured this variability, as demonstrated by F1-Score values exceeding 0.9. In contrast, the SP and PA classes presented the lowest F1-Score values, particularly for TempCNN and L-TAE. RF performed better in these classes, contributing to its superior OA results.

An analysis of the confusion matrices (Figure 7) reinforces that no significant misclassifications were observed for DC, SC and FV. However, PA thematic class was the main source of omissions errors across all models and thematic classes, particularly affecting SP classification, with omission errors of 10.06, 20.88 and 32.04 for RF, TempCNN and L-TAE, respectively. As shown in Figure 4, SP and PA had similar spectral patterns across all

Table 7. Overall Accuracy (OA) from each algorithm on the cross-temporal domain generalization test with ground samples. *indicates statistically significant difference compared to other models (p-value < 0.05).

	Random Forest (RF)	TempCNN	L-TAE
Overall Accuracy (OA)	0.917*	0.886*	0.859*

Table 8. F1-Score in each thematic class from the three algorithms on the cross-temporal domain generalization test with ground samples.

	Random Forest (RF)	TempCNN	L-TAE
Double Crop (DC)	0.972	0.960	0.944
Single Crop (SC)	0.918	0.914	0.903
Semi-perennial Crop (SP)	0.900	0.826	0.753
Pasture (PA)	0.850	0.801	0.766
Forest Vegetation (FV)	0.922	0.901	0.898

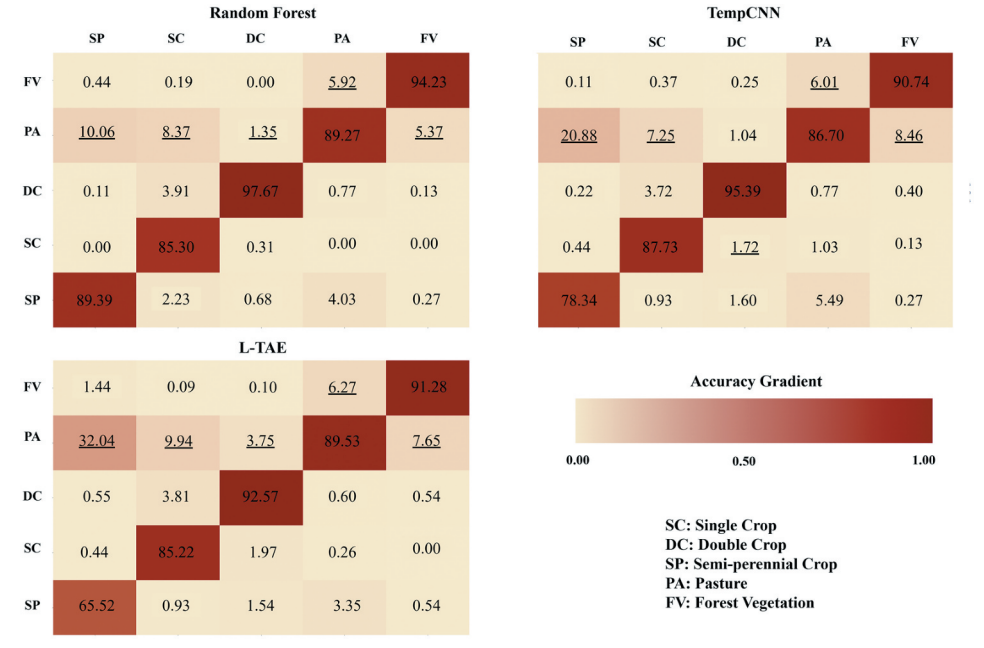


Figure 7. Confusion matrices showing the classification results from the test step, utilizing ground samples. The highest omission error for each thematic class is underlined.

spectral bands and vegetation indices, which contributed to misclassification and higher commission errors across other classes. Furthermore, Figures 5 and 6 highlight that PA samples exhibit spectral similarities with all thematic classes. Although the spectrotemporal patterns were distinct, the presence of mixed pixels likely introduced by MODIS spatial resolution adaptation may have led to misclassifications.

3.4.2. Applying the domain generalization for interannual classification

Given the demonstrated potential of the RF algorithm in our classification models, we proceeded to assess its interannual classification performance. This evaluation consisted of applying the RF model to time series data from the MODIS sensor to classify 18 crop years, thereby illustrating the cross-temporal DG approach for interannual classification tasks. For testing, four ground samples were selected in four municipalities (Figure 1). Figure 8 shows the classification results for each sample, with the highlighted segment in each temporal profile indicating the crop year for which ground references were available. Additionally, Table 9 presents the associated probability layers used to support the analysis and assess classification confidence. Values equal to or greater than 0.7 were highlighted to indicate the crop years in which the model demonstrated higher confidence in the classification results, while values below 0.7 suggest potential confusion between thematic classes.

Figure 8(a) illustrates a scenario where the model had difficulty distinguishing between the SP and PA classes. Initially, this location was used as SP for the 2016/2017 crop year during testing with ground samples, but the profile was ultimately classified as PA. The probability layer for this point indicates a classification probability of 0.40 for PA, reflecting the classifier's uncertainty in accurately categorizing this spectral pattern. However, for the profiles from the

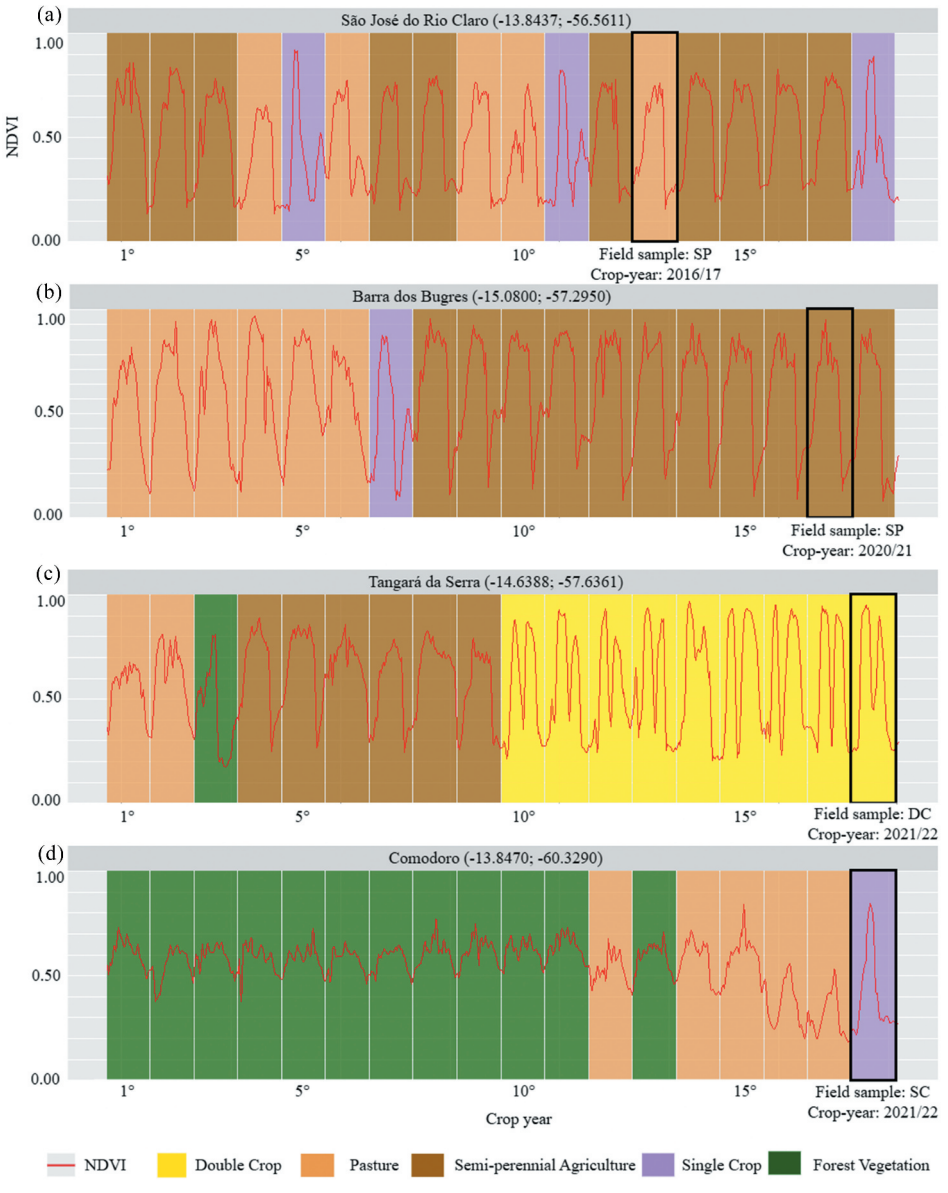


Figure 8. Interannual classification obtained with the cross-temporal Domain generalization of the Random Forest, highlighting the crop year corresponding to the ground sample (black squares) in the following municipalities of Mato Grosso state: (a) São José do Rio Claro (b), Barra dos Bugres, (c) Tangará da Serra and (d) Comodoro. SP: semi-perennial crop; SC: single crop; DC: double crop.

14th to 17th years, the classifier demonstrated higher classification confidence, with probabilities exceeding 0.88. Despite this improvement, subsequent periods within the same profile also revealed instances where the model faced challenges, as reflected in the probability layers.

Despite the classification challenges noted in Figure 8(a), the SP profiles in Figures 8(b,c) were successfully identified, demonstrating higher confidence in the probability layers. In

Table 9. Values of probability layers (PL) for each crop year across the 18-year time series classifications in Figure 8. PL values equal to or greater than 0.7 are underlined.

	São José do Rio Claro (Figure 8A)		Barra dos Bugres (Figure 8B)		Tangará da Serra (Figure 8C)		Comodoro (Figure 8D)	
Crop Year	Class	PL	Class	PL	Class	PL	Class	PL
1º	SP	0.534	PA	0.932	PA	0.956	FV	0.914
2º	SP	0.672	PA	0.880	PA	0.952	FV	0.644
3º	SP	0.600	PA	0.750	FV	0.426	FV	0.784
4º	PA	0.652	PA	0.556	SP	0.744	FV	0.796
5º	SC	0.522	PA	0.826	SP	0.730	FV	0.882
6º	PA	0.402	PA	0.998	SP	0.542	FV	0.758
7º	SP	0.470	SC	0.432	SP	0.788	FV	0.832
8º	SP	0.546	SP	0.828	SP	0.544	FV	0.938
9º	PA	0.474	SP	0.912	SP	0.710	FV	1.000
10º	PA	0.416	SP	0.896	DC	0.784	FV	0.994
11º	SC	0.328	SP	0.944	DC	0.724	FV	1.000
12º	SP	0.620	SP	0.892	DC	0.634	PA	0.556
13º	PA	0.400	SP	0.898	DC	0.786	FV	0.756
14º	SP	0.884	SP	0.904	DC	0.700	PA	0.772
15º	SP	0.914	SP	0.936	DC	0.99	PA	0.616
16º	SP	0.966	SP	0.994	DC	0.998	PA	0.902
17º	SP	0.918	SP	0.868	DC	0.926	PA	0.876
18º	SC	0.620	SP	0.536	DC	0.978	SC	0.924

Single Crop (SC) – Semi-perennial Crop (SP) – Double Crop (DC) – Pasture (PA) – Forest Vegetation (FV) – Probability Layer (PL)

Figure 8(b), both PA and SP profiles were classified with probabilities exceeding 0.80, and in some instances, surpassing 0.90. Moreover, a classification challenge was found in the 7th crop year, likely due to a LULC transition. Similarly, in Figure 8(c), during the 3rd crop year presented a potential crop transition that posed a classification challenge, as indicated by a lower probability layer of 0.426. Despite this, the other transitions between SP and DC profiles were effectively identified.

Finally, Figure 8(d) illustrates the LULC transition between the FV and PA classes. The probability layers revealed no inconsistencies, except during the 12th crop year, where a probability of 0.556 was noted for the PA class. Additionally, this profile highlighted the 18th crop year, during which this point was classified as belonging to SC class during testing. This transition exemplifies the dynamic changes between natural vegetation and agricultural cropland, as identified through the interannual classification.

4. Discussion

4.1. Training samples and SOM analysis

The TerraClass mapping system, originally developed in Brazil, aims to monitor changes in natural vegetation and agricultural land cover. Our approach demonstrated its potential in mitigating the scarcity of training samples by retrieving a dataset of 40,000 samples for five thematic classes across the Brazilian state of Mato Grosso (903,357 km^2), specifically for the 2019/2020 crop year. Our sample collection methodology included an adaptation process to align with MODIS 250-metres spatial resolution and employed a spatial erosion to reduce the impact of mixed pixels derived from this adaption.

In RS studies, collecting training samples based on existing LULC maps is a common approach. For instance, Ge et al. (2021), Hao et al. (2020) and Massey et al. (2017) utilized

samples derived from the Cropland Data Layer to train classification algorithms. Furthermore, Hao et al. (2020) emphasized the importance of this approach in model adaption methodology, noting that their TD dataset had less than 1000 samples. In contrast, our approach leveraged TerraClass maps to generate a larger training dataset (40000), which effectively captured intra-class variability, especially for SC, SP and DC.

As shown in Figure 6, SC exhibited distinct NDVI patterns, consistent with findings from Zheng et al. (2022), which reported spectral differences between cotton and soybean fields within the SC cultivation. Furthermore, other minor intra-class variability was identified, such as three cycles of crop in DC class. These findings underscore the capability of TerraClass maps to generate large and reliable samples sets for remote sensing applications. However, some limitations of TerraClass should be considered: (I) thematic classes are not defined at the crop level; (II) LULC classification maps inherently carry a degree of uncertainty; and (III) there is a need for an adaption process when using sensor with spatial resolution different from that of Sentinel 2/MSI, which may introduce mixed pixels into the samples. In this study, we proposed an adaptation process to MODIS data and applied SOM to address uncertainties associated with classification and reduce inclusion of mixed pixels. Nonetheless, future studies may explore the use of TerraClass for crop level classification and further refine the adaptation process.

SOM analysis revealed three key findings: (I) notable similarity between the spectral patterns of PA and SP classes; (II) significant intra-class variability, especially within the SC class; and (III) presence of samples with inconsistent patterns for their respective thematic class, and posterior exclusion of 8.188 inconsistent samples. The SOM was applied to samples distributed across Mato Grosso State. Similar results were reported by L. A. Santos, Ferreira, Picoli, et al. (2021) and L. A. Santos, Ferreira, Camara, et al. (2021), who applied SOM to the entire Cerrado biome. These findings underscore the potential of SOM in handling with time series data allowing a systematic and scalable approach to sample characterization. Although our approach was based on generic thematic classes, future studies focusing on crop-level classification or larger study areas could benefit from using SOM for more detailed sample assessment and refinement.

4.2. Interpretation of classification results

While the classification models presented an OA greater than 0.960 for the 2019/2020 crop year, their performance during the test step highlighted the inherent challenges of the DG approach. In the test, the classification models achieved an OA of 0.917 for RF, 0.886 for TempCNN and 0.859 for L-TAE. Although a performance loss was observed, the ground samples for testing were collected during field campaigns and in three different crop years (2016/2017, 2020/2021 and 2021/2022). These samples represent independent data from those used for training and validation, which were based from the same crop year (2019/2020) and derived from TerraClass maps. Consequently, the observed performance loss was not attributed to overfitting, as the training process did not include samples from other crop years.

Our approach can be compared with the studies of Massey et al. (2017) and Luciano, Campagnuci, and le Maire (2022), which also successfully applied cross-temporal DG using RF. Our results outperformed their reported OA of 0.78 and 0.82, respectively. However, it is important to note that the ground samples used for testing in our study

presented unbalanced data, as shown in Table 4. While the results and interpretations support the model generalization capability, interpretations regarding a specific crop year require more data. Furthermore, Massey et al. (2017) conducted DG across a broader area, while Luciano, Campagnuci, and le Maire (2022) employed a 33-year time series, allowing greater temporal and spatial generalizations. Future studies may further explore the scalability of DG approaches by incorporating longer time series, wider spatial domains and more balanced reference data to assess specific crop years. In this study, interannual classification was applied for 18 years of MODIS SITS data; however, recent crop years, such as 2022/2023 onward, remain to be investigated. Additionally, the Pantanal biome in Mato Grosso was not included. Our future studies may consider expanding the application of the classification models to other biomes or states across Brazil, thereby enhancing the generation of LULC maps over multiple years and regions throughout the country.

Regarding the performance of each algorithm, all three classifiers achieved F1-Scores above 0.9 and low omission errors for the SC, DC, and FV thematic classes. However, higher omissions errors were observed for SP, particularly with TempCNN and L-TAE. The confusion matrices highlighted that the PA class was the primary source of misclassifications across all thematic classes. Among the models, RF demonstrated better adaptation to the PA and achieved better results. Zheng et al. (2022) also reported the challenges in distinguishing sugarcane and pasture areas in Mato Grosso due to the similarity of their NDVI patterns. According to the authors, vegetation in Mato Grosso decreases during dry season (August to October) and increases afterwards, which is similar to the harvest and sowing periods of sugarcane in Mato Grosso. Their study achieved omission, commission and OA of 5.65, 12.93 and 91,47, respectively. However, their classification considered two thematic classes (sugarcane and non-sugarcane) whereas our RF model achieved an omission error of 10.06% while classifying five thematic classes, highlighting its robustness in a more complex classification scenario.

Regarding the results of TempCNN and L-TAE, the robustness of DL algorithms in handling SITS data was previously highlighted (Fu et al. 2024; Pelletier, Webb, and Petitjean 2019). In a spatiotemporal DG approach, Feng et al. (2024) employed Synthetic Aperture Radar (SAR) data and achieved an OA of 96.54 using attention-based DL algorithm, outperforming RF, which reached 82.65. Their study demonstrated the potential of combining SAR data with DL techniques to expand the applicability of DG. In contrast, our approach reached better performance using RF. However, the difference can be attributed to the characteristics of our dataset, which is based on MODIS images with 250 m spatial resolution, four spectral band (BLUE-NIR1-NIR2-RED) and two vegetation indices (NDVI and EVI). Additionally, the presence of mixed pixels from TerraClass maps adaptation may have introduced noise in the training samples, adding challenges to the training process. Despite the robustness of TempCNN and L-TAE, the data provided in training may not be enough for capturing significant differences between PA and SP classes. Previous studies applying DL algorithm to complex LULC tasks such as Integrated Livestock Systems (Bueno et al. 2023; Werner et al. 2024) and crop-level classification classes (C. L. M. D. O. Santos et al. 2019; Souza et al. 2020), successfully leveraged DL in combination with a broader set of vegetation indices and higher spatial resolution data from sensors, such as Landsat 8/OLI and Sentinel-2/MSI.

Furthermore, previous DG and Domain Adaptation approaches have utilized Landsat data collections with 30 m spatial resolution, offering more detailed data for classification models (Mohammadi, Belgiu, and Stein 2024; Luciano, Campagnuci, and le Maire 2022). Future applications of DG may benefit from incorporating additional vegetation indices in combination with DL algorithms, enhancing their capacity to handle more complex scenarios and improving the discrimination of similar thematic classes.

In tropical regions such as Brazil, cloud cover presents a challenge to remote sensing analyses by introducing gaps in the time series and hindering the recognition of spectrotemporal patterns (Prudente et al. 2020). In areas and time periods with high cloud incidence, the integration of multiple data sources such as Sentinel-2/MSI or Landsat 8/OLI can support the generation of cloud-free time series and provide valuable information for classification models (Whitcraft et al. 2015). In this study, cloud cover was mitigated using MODIS pixel quality products and bilinear interpolation to fill temporal gaps, thereby supporting the performance of the classification models. Future studies could address cloud-cover related limitations incarnating SAR data, which offer robust and cloud-free data for crop classification, as demonstrated by Feng et al. (2024).

Regarding the interannual classification of the 18-year time series, and in comparison, to the approach of Luciano, Campagnuci, and le Maire (2022), our evaluation was based on the analyses of the probability layers, which provided valuable insights into how the model classified each crop year throughout the time series. Specifically, in observed transition between thematic classes, these layers enabled a deeper understanding on model behaviour for future improvements. However, this evaluation was limited to four ground samples points selected to visualize and assess specific spectrotemporal patterns such as LULC transitions between SP and DC (Figure 8(c)), complementing the results from the ground samples for test (Table 4). To further assess these results, future studies could expand this analysis by classifying additional locations and comparing model outputs with other available reference maps, as demonstrated by Luciano, Campagnuci, and le Maire (2022) and Massey et al. (2017), including datasets such as MapBiomas or labelled samples from independent studies (Del'Arco Sanches et al. 2018).

Despite some limitations identified, our results demonstrated that RF model successfully exhibited generalization capability, enabling the classification of an 18-year period of MODIS SITS data. Although classification errors were observed in specific crop years, primarily due to the training set characteristics and the inherent challenges associated in transition areas, the RF model effectively detected LULC changes over time and accurately classified spectrotemporal profiles associated across five thematic classes. Those results align with the objectives of this study and support our hypothesis that a model trained on a robust and well-characterized dataset can maintain its performance across the MODIS time series, highlighting the potential of our cross-temporal DG approach.

5. Conclusion

This study presents a novel method for interannual classification of MODIS time series data, leveraging the cross-temporal Domain Generalization approach. Although TerraClass data originally aligns with Sentinel 2/MSI at 10 m spatial resolution, our adaptation to MODIS

250 m resolution provided a robust and reliable data source for DG. Additionally, sample inconsistencies and intra-class variability were identified using SOM algorithm, enabling a consistent and well-structured sample characterization process. Our findings indicate that SOM played a crucial role in the overall success of the cross-temporal DG implementation.

The RF classification model achieved an OA of 0.917, outperforming two DL algorithms, TempCNN and L-TAE, particularly in distinguishing SP and PA thematic classes. Despite its simpler architecture, RF effectively adapted to the variability within the samples and demonstrated strong generalization capability when applied to data from different crop years. These results demonstrated the applicability of cross-temporal DG for classifying crop years that lack labelled samples, thus contributing to the expansion of LULC analysis over time. As a next step, we aim to evaluate the scalability of this approach across recent crop years, such as 2022/2023 onward, and other Brazilian states within the Cerrado and Amazon biomes, and to investigate its application in additional regions, such as the Pantanal biome.

Our findings provide valuable insights and direction for future studies employing DG in SITS classification, helping to mitigate the challenges posed by sample scarcity. Moreover, future studies may explore the integration of higher-resolution SITS data, a broader set of vegetation indices and the inclusion of SAR data to address issues related to cloud cover and mixed pixel. These enhancements, when combined with deep learning algorithms, could further strengthen the applicability of DG in complex scenarios and support the development of more detailed thematic classifications.

Acknowledgments

The authors express their gratitude to the School of Agricultural Engineering (FEAGRI) at the University of Campinas (UNICAMP) for providing infrastructure, and to the Brazilian Agricultural Research Corporation (Embrapa) for supplying ground samples used in this study.

Disclosure statement

The authors declare that the research was conducted in the absence of any commercial or financial relationship that could be construed as a potential conflict of interest.

Funding

This research was funded by the National Council for Scientific and Technological Development (CNPq) [Grant: 130329/2022-3]; Conselho Nacional de Desenvolvimento Científico e Tecnológico [130329/2022-3].

ORCID

Daniel H. Shibuya http://orcid.org/0009-0005-1166-3517
Júlio C.D.M. Esquerdo http://orcid.org/0000-0001-7190-2931
João P.S. Werner http://orcid.org/0000-0001-5219-3551

Data availability statement

Data will be made available on request

References

Almeida, C. A. D., A. C. Coutinho, J. C. D. M. Esquerdo, M. Adami, A. Venturieri, C. Guerreiro Diniz, N. Dessay, L. Durieux, and A. R. Gomes. 2016. "High Spatial Resolution Land Use and Land Cover Mapping of the Brazilian Legal Amazon in 2008 Using landsat-5/tm and MODIS Data." *Acta Amazonica* 46 (3). Instituto Nacional de Pesquisas da Amazonia: 291–302. https://doi.org/10.1590/1809-4392201505504.

Arvor, D., M. Meirelles, V. Dubreuil, A. Bégué, and Y. E. Shimabukuro. 2012. "Analyzing the Agricultural Transition in Mato grosso, Brazil, Using Satellite-Derived Indices." *Applied Geography* 32 (2). Elsevier Ltd: 702–713. https://doi.org/10.1016/j.apgeog.2011.08.007.

Belgiu, M., and L. Drăgu. 2016. "Random Forest in Remote Sensing: A Review of Applications and Future Directions." *ISPRS Journal of Photogrammetry & Remote Sensing* 114:24–31. https://doi.org/10.1016/j.isprsjprs.2016.01.011.

Blanchard, G., G. Lee, and C. Scott. 2011. "Generalizing from Several Related Classification Tasks to a New Unlabeled Sample" 25th International Conference on Neural Information Processing Systems (NIPS'11) 2178–2186. https://dl.acm.org/doi/10.5555/2986459.2986702."

Breiman, L. 2001. "Random Forests." *Machine Learning* 45 (1): 5–32. https://doi.org/10.1023/A:1010933404324.

Brown, J. C., J. H. Kastens, A. Camargo Coutinho, D. de Castro Victoria, and C. R. Bishop. 2013. "Classifying Multiyear Agricultural Land Use Data from Mato Grosso Using Time-Series MODIS Vegetation Index Data." *Remote Sensing of Environment* 130 (March): 39–50. https://doi.org/10.1016/j.rse.2012.11.009.

Bueno, I. T., J. F. G. Antunes, A. A. Dos Reis, J. P. S. Werner, A. P. S. G. D. D. Toro, G. K. D. A. Figueiredo, J. C. D. M. Esquerdo, R. A. C. Lamparelli, A. C. Coutinho, and P. S. G. Magalhães. 2023. "Mapping Integrated Crop-Livestock Systems in Brazil with Planetscope Time Series and Deep Learning." *Remote Sensing of Environment* 299 (December). Elsevier Inc: 113886. https://doi.org/10.1016/j.rse.2023.113886.

Companhia Nacional de Abastecimento (CONAB) [National Supply Company]. 2021. *Calendário Agrícola (plantio e colheita)*, Brasília: CONAB. https://www.conab.gov.br.

Congalton, R. G. 1991. "A Review of Assessing the Accuracy of Classifications of Remotely Sensed Data." *Remote Sensing of Environment* 37 (1): 35–46. https://doi.org/10.1016/0034-4257(91)90048-B.

Dantas Chaves, M. E., and M. D. C. Alves. 2019. "Recent Applications of the MODIS Sensor for Soybean Crop Monitoring and Deforestation Detection in Mato Grosso, Brazil." *CAB Reviews: Perspectives in Agriculture, Veterinary Science, Nutrition & Natural Resources* 14:1–9. February. https://doi.org/10.1079/PAVSNNR201914007.

Del'Arco Sanches, I., R. Q. Feitosa, P. M. A. Diaz, M. D. Soares, A. J. B. Luiz, B. Schultz, and L. E. P. Maurano. 2018. "Campo Verde Database: Seeking to Improve Agricultural Remote Sensing of Tropical Areas." *IEEE Geoscience & Remote Sensing Letters* 15 (3). Institute of Electrical and Electronics Engineers Inc.: 369–373. https://doi.org/10.1109/LGRS.2017.2789120.

Esquerdo, J. C. D. M., J. F. G. Antunes, A. C. Coutinho, E. A. Speranza, A. A. Kondo, and J. L. dos Santos. 2020. "SATVeg: A Web-Based Tool for Visualization of MODIS Vegetation Indices in South America." *Computers and Electronics in Agriculture* 175 (June 2019). Elsevier: 105516. https://doi.org/10.1016/j.compag.2020.105516.

Feng, L., D. Gui, S. Han, T. Qiu, and Y. Wang. 2024. "Integrating Optical and SAR Time Series Images for Unsupervised Domain Adaptive Crop Mapping." *Remote Sensing* 16 (8). Multidisciplinary Digital Publishing Institute (MDPI): 1464. https://doi.org/10.3390/rs16081464.

Fu, Y., Z. Zhu, L. Liu, W. Zhan, T. He, H. Shen, J. Zhao, et al. 2024. "Remote Sensing Time Series Analysis: A Review of Data and Applications." *Journal of Remote Sensing* 4. January. American Association for the Advancement of Science (AAAS). https://doi.org/10.34133/remotesensing.0285.

Garnot, V. S. F., and L. Landrieu. 2020. "Lightweight Temporal Self-Attention for Classifying Satellite Images Time Series." *Lecture Notes in Computer Science (including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics) 12588 LNAI*: 171–181. https://doi.org/10.1007/978-3-030-65742-0_12.

Ge, S., J. Zhang, Y. Pan, Z. Yang, and S. Zhu. 2021. "Transferable Deep Learning Model Based on the Phenological Matching Principle for Mapping Crop Extent." *International Journal of Applied Earth Observation and Geoinformation* 102 (October). Elsevier B.V.: 102451. https://doi.org/10.1016/j.jag.2021.102451.

Gerald, B. 2018. "A Brief Review of Independent, Dependent and One Sample t-Test." *International Journal of Applied Mathematics and Theoretical Physics* 4 (2). Science Publishing Group: 50. https://doi.org/10.11648/j.ijamtp.20180402.13.

Hao, P., L. Di, C. Zhang, and L. Guo. 2020. "Transfer Learning for Crop Classification with Cropland Data Layer Data (CDL) as Training Samples." *Science of the Total Environment* 733. Elsevier B.V.: 138869. 138869. https://doi.org/10.1016/j.scitotenv.2020.138869.

Huete, A. R., H. Liu, and J.D.V.L. Wim. 1997. "The Use of Vegetation Indices in Forested Regions: Issues of Linearity and Saturation." In Proceedings of IGARSS'97 – 1997 IEEE International Geoscience and Remote Sensing Symposium: Remote Sensing – A Scientific Vision for Sustainable Development, vol. 4, 1966–1968. IEEE. https://doi.org/10.1109/IGARSS.1997.609169.

Ismail Fawaz, H., G. Forestier, J. Weber, L. Idoumghar, and P. Alain Muller. 2019. "Deep Learning for Time Series Classification: A Review." *Data Mining and Knowledge Discovery* 33 (4). Springer US: 917–963. https://doi.org/10.1007/s10618-019-00619-1.

Johnson, D. M. 2019. "Using the Landsat Archive to Map Crop Cover History Across the United States." *Remote Sensing of Environment* 232 (July). Elsevier: 111286. https://doi.org/10.1016/j.rse.2019.111286.

Justice, C. O., J. R. G. Townshend, E. F. Vermote, and E. Mauoka. 2002. "An Overview of MODIS Land Data Processing and Product Status." *Remote Sensing of Environment* 83 (1–2): 3–15. https://doi.org/10.1016/S0034-4257(02)00084-6.

Khanna, S., B. Wallace, K. Bala, and B. Hariharan. 2022. "Activation Regression for Continuous Domain Generalization with Applications to Crop Classification," April. http://arxiv.org/abs/2204.07030.

Kingma, D. P., and J. Ba. 2014. "Adam: A Method for Stochastic Optimization." *December*. http://arxiv.org/abs/1412.6980.

Kohonen, T. 1990. "The Self-Organizing Map." *Proceedings of the IEEE* 78 (9): 1464–1480. https://doi.org/10.1109/5.58325.

Llano, X. 2023. *SMByC-Ideam. AcATaMa - QGIS Plugin for Accuracy Assessment of Thematic Maps, Version 23.4*. https://github.com/SMByC/AcATaMa.

Luciano, A. C. D. S., B. C. G. Campagnuci, and G. le Maire. 2022. "Mapping 33 Years of Sugarcane Evolution in São Paulo State, Brazil, Using Landsat Imagery and Generalized Space-Time Classifiers." *Remote Sensing Applications: Society & Environment* 26 (April). Elsevier B.V.: 100749. https://doi.org/10.1016/j.rsase.2022.100749.

Luciano, A. C. D. S., M. C. A. Picoli, J. Vieira Rocha, H. Coutinho Junqueira Franco, G. Martineli Sanches, M. Regis Lima Verde Leal, and G. le Maire. 2018. "Generalized Space-Time Classifiers for Monitoring Sugarcane Areas in Brazil." *Remote Sensing of Environment* 215 (October 2017). Elsevier: 438–451. https://doi.org/10.1016/j.rse.2018.06.017.

Massey, R., T. T. Sankey, R. G. Congalton, K. Yadav, P. S. Thenkabail, M. Ozdogan, and A. J. Sánchez Meador. 2017. "MODIS Phenology-Derived, Multi-Year Distribution of Conterminous U.S. Crop Types." *Remote Sensing of Environment* 198 (September). Elsevier Inc.: 490–503. https://doi.org/10.1016/j.rse.2017.06.033.

Mohammadi, S., M. Belgiu, and A. Stein. 2024. "A source-Free Unsupervised Domain Adaptation Method for Cross-Regional and Cross-Time Crop Mapping from Satellite Image Time Series." *Remote Sensing of Environment* 314 (December). Elsevier Inc: 114385. https://doi.org/10.1016/j.rse.2024.114385.

Oldoni, L. V., I. Del', A. Sanches, V. Hugo, R. Prudente, D. Corte Vieira, and F. Furlan Gama. 2019. "caracterização da dinâmica da soja, milho e algodão com base em dados SAR polarimétricos do Sentinel-1A." https://www.modis.cnptia.embrapa.br/.

Olofsson, P., G. M. Foody, M. Herold, S. V. Stehman, C. E. Woodcock, and M. A. Wulder. 2014. "Good Practices for Estimating Area and Assessing Accuracy of Land Change." *Remote Sensing of Environment. Elsevier Inc* 148:42–57. https://doi.org/10.1016/j.rse.2014.02.015.

Pelletier, C., S. Valero, J. Inglada, N. Champion, and G. Dedieu. 2016. "Assessing the Robustness of Random Forests to Map Land Cover with High Resolution Satellite Image Time Series Over Large Areas." *Remote Sensing of Environment* 187. Elsevier Inc.: 156–168. https://doi.org/10.1016/j.rse.2016.10.010.

Pelletier, C., G. I. Webb, and F. Petitjean. 2019. "Temporal Convolutional Neural Network for the Classification of Satellite Image Time Series." *Remote Sensing* 11 (5): 523–525. https://doi.org/10.3390/rs11050523.

Picoli, M. C. A., G. Camara, I. Sanches, R. Simões, A. Carvalho, A. Maciel, A. Coutinho, et al. 2018. "Big Earth Observation Time Series Analysis for Monitoring Brazilian Agriculture." *Isprs Journal of Photogrammetry & Remote Sensing* 145 (August). Elsevier: 328–339. https://doi.org/10.1016/j.isprsjprs.2018.08.007.

Prudente, V. H. R., V. Martins, D. Corte Vieira, N. R. de França e Silva, M. Adami, and I. Del Arco Sanches. 2020. "Limitations of Cloud Cover for Optical Remote Sensing of Agricultural Areas Across South America." *Remote Sensing Applications: Society & Environment* 20:100414. (November). Elsevier B.V. https://doi.org/10.1016/j.rsase.2020.100414.

Rouse, J. W., R. H. Haas, J. A. Schell, and D. W. Deering. 1973. "Monitoring Vegetation Systems in the Great Plains with ERTS." *Proceedings of the Third Earth Resources Technology Satellite-1 Symposium*, vol. 1. Washington, D.C.: 309–317. NASA.

Santos, C. L. M. D. O., R. Augusto Camargo Lamparelli, G. Kelly Dantas Araújo Figueiredo, S. Dupuy, J. Boury, A. C. D. S. Luciano, R. da Silva Torres, and G. le Maire. 2019. "Classification of Crops, Pastures, and Tree Plantations Along the Season with Multi-Sensor Image Time Series in a Subtropical Agricultural Region." *Remote Sensing* 11 (3): 1–27. https://doi.org/10.3390/rs11030334.

Santos, L. A., K. Ferreira, G. Camara, M. C. A. Picoli, and R. E. Simoes. 2021. "Quality Control and Class Noise Reduction of Satellite Image Time Series." *ISPRS Journal of Photogrammetry & Remote Sensing* 177 (July 2020). Elsevier B.V.: 75–88. https://doi.org/10.1016/j.isprsjprs.2021.04.014.

Santos, L. A., K. Ferreira, M. Picoli, G. Camara, R. Zurita-Milla, and E. W. Augustijn. 2021. "Identifying Spatiotemporal Patterns in Land Use and Cover Samples from Satellite Image Time Series." *Remote Sensing* 13 (5): 1–21. https://doi.org/10.3390/rs13050974.

Santosh, K. C., N. Das, and S. Ghosh. 2022. "Deep Learning: A Review." *Deep Learning Models for Medical Imaging*. https://doi.org/10.1016/b978-0-12-823504-1.00012-x.

Simoes, R., G. Camara, G. Queiroz, F. Souza, P. R. Andrade, L. Santos, A. Carvalho, and K. Ferreira. 2021. "Satellite Image Time Series Analysis for Big Earth Observation Data." *Remote Sensing* 13 (13): 1–20. https://doi.org/10.3390/rs13132428.

Simoes, R., M. C. A. Picoli, G. Camara, A. Maciel, L. Santos, P. R. Andrade, A. Sánchez, K. Ferreira, and A. Carvalho. 2020. "Land Use and Cover Maps for Mato Grosso State in Brazil from 2001 to 2017." *Scientific Data* 7 (1): 1–10. https://doi.org/10.1038/s41597-020-0371-4.

Souza, C. M., J. Z. Shimbo, M. R. Rosa, L. L. Parente, A. A. Alencar, B. F. T. Rudorff, H. Hasenack, et al. 2020. "Reconstructing Three Decades of Land Use and Land Cover Changes in Brazilian Biomes with Landsat Archive and Earth Engine." *Remote Sensing* 12 (17). MDPI AG. 2735. https://doi.org/10.3390/RS12172735.

Vieira, D. C., I. Del Arco Sanches, B. Montibeller, V. Hugo Rohden Prudente, M. C. Hansen, A. Baggett, and M. Adami. 2022. "Cropland Expansion, Intensification, and Reduction in Mato Grosso State, Brazil, Between the Crop Years 2000/01 to 2017/18." *Remote Sensing Applications: Society & Environment* 28 (August). Elsevier B.V.: 100841. https://doi.org/10.1016/j.rsase.2022.100841.

Wang, J., M. Bretz, M. A. A. Dewan, and M. A. Delavar. 2022. "Machine Learning in Modelling Land-Use and Land cover-Change (LULCC): Current Status, Challenges and Prospects." *Science of the Total Environment* 822:153559. https://doi.org/10.1016/j.scitotenv.2022.153559.

Wang, J., C. Lan, C. Liu, Y. Ouyang, T. Qin, W. Lu, Y. Chen, W. Zeng, and P. S. Yu. 2021. "Generalizing to Unseen Domains: A Survey on Domain Generalization." March. http://arxiv.org/abs/2103.03097.

Wang, Y., Y. Sun, X. Cao, Y. Wang, W. Zhang, and X. Cheng. 2023. "A Review of Regional and Global Scale Land use/Land Cover (LULC) Mapping Products Generated from Satellite Remote Sensing." *Isprs Journal of Photogrammetry & Remote Sensing*. Elsevier B.V. https://doi.org/10.1016/j.isprsjprs.2023.11.014.

Werner, J. P. S., M. Belgiu, I. T. Bueno, A. A. Dos Reis, A. P. S. G. D. Toro, J. F. G. Antunes, A. Stein, et al. 2024. "Mapping Integrated Crop–Livestock Systems Using Fused Sentinel-2 and planetscope Time Series and Deep Learning." *Remote Sensing* 16 (8). Multidisciplinary Digital Publishing Institute (MDPI): 1421. https://doi.org/10.3390/rs16081421.

Whitcraft, A. K., I. Becker-Reshef, B. D. Killough, and C. O. Justice. 2015. "Meeting Earth Observation Requirements for Global Agricultural Monitoring: An Evaluation of the Revisit Capabilities of Current and Planned Moderate Resolution Optical Earth Observing Missions." *Remote Sensing* 7 (2). MDPI AG: 1482–1503. https://doi.org/10.3390/rs70201482.

Xu, J., Y. Zhu, R. Zhong, Z. Lin, J. Xu, H. Jiang, J. Huang, H. Li, and T. Lin. 2020. "Deepcropmapping: A Multi-Temporal Deep Learning Approach with Improved Spatial Generalizability for Dynamic Corn and Soybean Mapping." *Remote Sensing of Environment* 247 (March). Elsevier: 111946. https://doi.org/10.1016/j.rse.2020.111946.

Zhang, M., G. Abrahao, A. Cohn, J. Campolo, and S. Thompson. 2021. "A Modis-Based Scalable Remote Sensing Method to Estimate Sowing and Harvest Dates of Soybean Crops in Mato Grosso, Brazil." *Heliyon* 7 (7). Elsevier Ltd: e07436. https://doi.org/10.1016/j.heliyon.2021.e07436.

Zheng, Y., A. Cláudia Dos Santos Luciano, J. Dong, and W. Yuan. 2022. "High-Resolution Map of Sugarcane Cultivation in Brazil Using a Phenology-Based Method." *Earth System Science Data* 14 (4). Copernicus Publications: 2065–2080. https://doi.org/10.5194/essd-14-2065-2022.

Zhong, L., P. Gong, and G. S. Biging. 2014. "Efficient Corn and Soybean Mapping with Temporal Extendability: A Multi-Year Experiment Using Landsat Imagery." *Remote Sensing of Environment* 140 (January): 1–13. https://doi.org/10.1016/j.rse.2013.08.023.

Zhou, K., Z. Liu, Y. Qiao, T. Xiang, and C. Change Loy. 2023. "Domain Generalization: A Survey." IEEE Transactions on Pattern Analysis and Machine Intelligence 45 (4): 4396–4415. https://doi.org/10.1109/TPAMI.2022.3195549.

Zhu, S., C. Wu, B. Du, and L. Zhang. 2023. "Style and Content Separation Network for Remote Sensing Image Cross-Scene Generalization." *ISPRS Journal of Photogrammetry & Remote Sensing* 201 (July). Elsevier B.V.: 1–11. https://doi.org/10.1016/j.isprsjprs.2023.05.007.

Appendices

A. Intra-class Variability

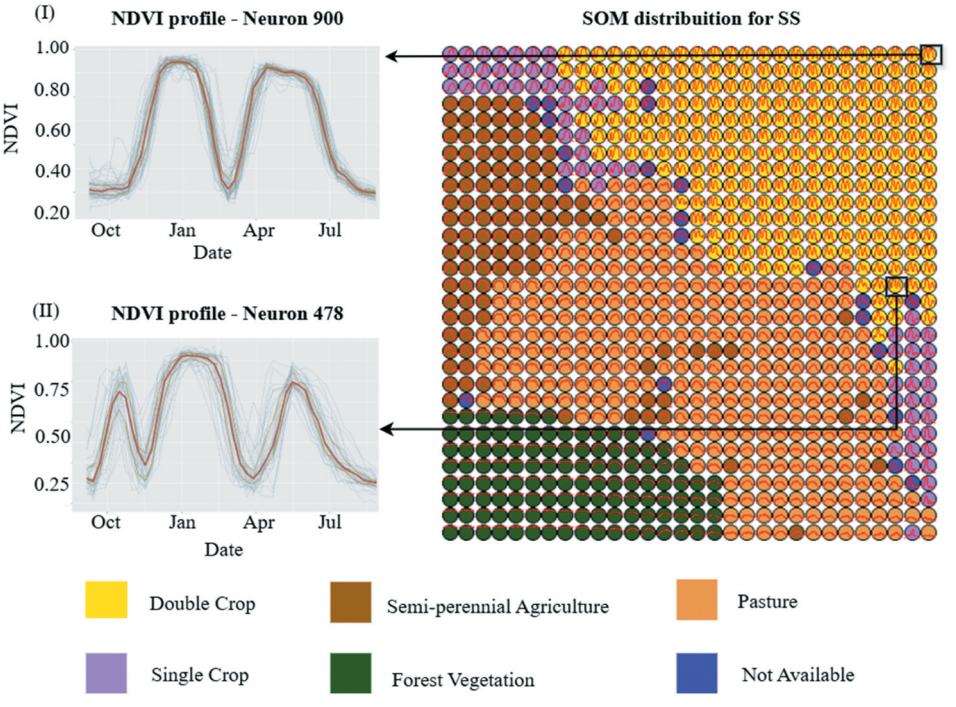


Figure A1. Self-Organizing-maps (SOM) of SOM sampling (SS), highlighting two examples of NDVI profiles from neurons 900 (I) and 478 (II), that represents two distinct groups of double crop (DC) thematic class.

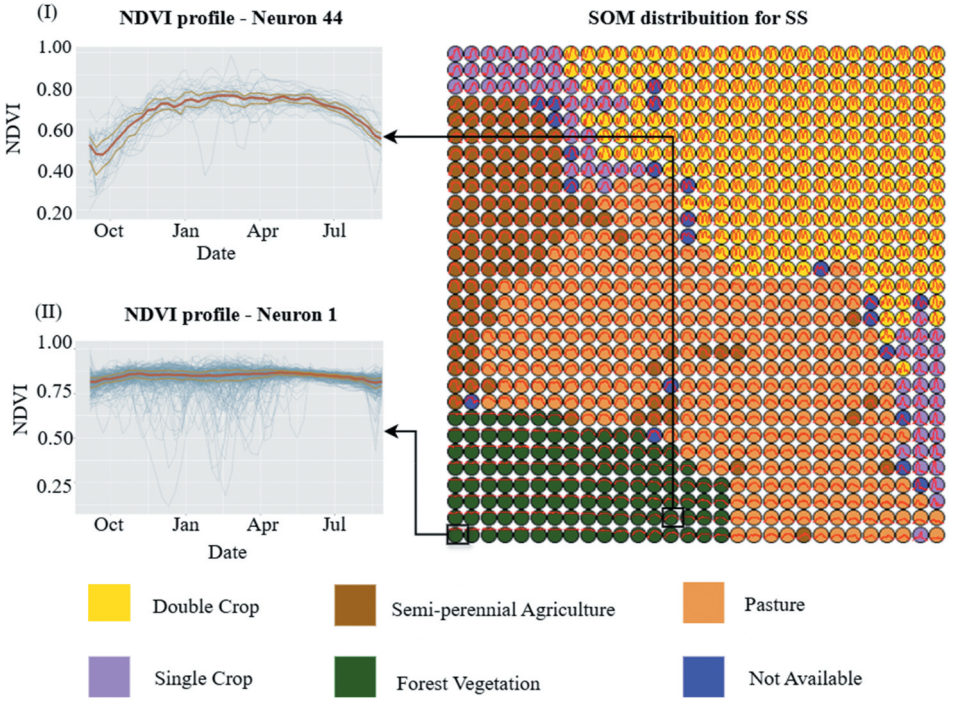


Figure A2. Self-Organizing-maps (SOM) of SOM sampling (SS), highlighting two examples of NDVI profiles from neurons 44 (I) and 1 (II), that represents two distinct groups of Forest vegetation (FV) thematic class.

B. *Base model F1-score and confusion matrices*

Table B1. F1-Score in each thematic class from the three algorithms on the base model validation.

	Random Forest (RF)	TempCNN	L-TAE
Double Crop (DC)	0.998	0.996	0.980
Single Crop (SC)	0.994	0.982	0.923
Semi-perennial Crop (SP)	0.984	0.942	0.913
Pasture (PA)	0.989	0.966	0.941
Forest Vegetation (FV)	0.998	0.996	0.992

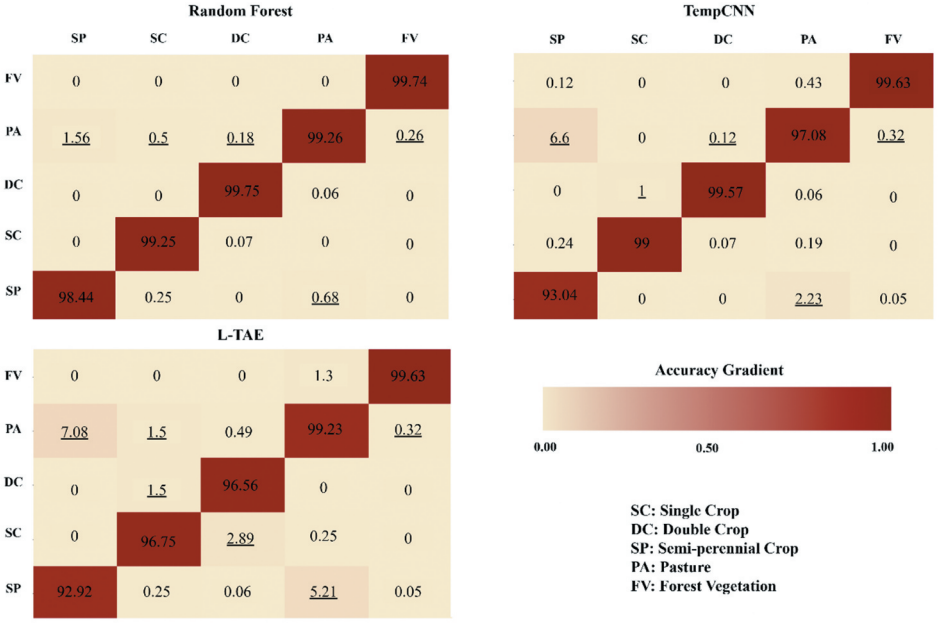


Figure B1. Confusion matrices showing the classification results from validation step. The highest omission error for each thematic class is underlined.